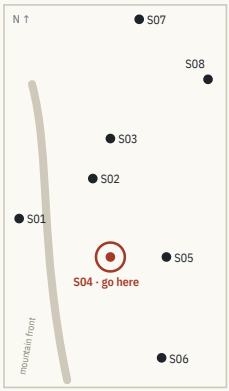
RIDGELINE SCHOOLS WEATHER NETWORK

Is the spring data usable, and where should the technician go?

Data: 1 March – 11 April 2025
8 stations, readings every 10 min
For the network coordinator

Question 2 · Where should the technician go next week?

Send them to S04 Foothills Community Garden.



Its thermometer reads too warm in sunshine, and the error grows every week: daily highs went from about 1 °C too warm in late March to about 3–4 °C by mid-April. This cannot be fixed afterwards in the data. Only a visit stops it.

1. **First, before touching anything:** in sunshine, hold a handheld thermometer in shade beside the shield for 5 minutes; note both readings. This confirms the cause.
2. **Then the shield** (white stack of plates around the sensor): the 15 March repaint of the post may have put paint in its vents. Clean the vents or fit a spare shield.
3. **Rain gauge:** check the funnel is still clear after the 2 April clean-out and fit a leaf guard.

> **Why not S03?** Its thermometer is failing too (it froze for three days and sometimes sends impossible values), but in between it reads correctly and the bad values are easy to remove. A volunteer can swap in a spare probe. S04's error hides inside normal-looking numbers, so it does more harm.

Question 1 · Which data is good enough for the spring summary?

Nearly all of it. Leave out the red parts below.

We found 11 real faults. Two cannot be repaired (red below). We have already corrected or removed the rest. No station needs to be dropped.

Do not use for the summary:

- **S04 daily maximum temperatures** from late March onwards (night-time lows are fine).
- **S04 day-by-day rain**, 23 March to 2 April. The blocked gauge released all the stored water at once when cleaned, so its *six-week total is still right* (33 mm, neighbours 28–45 mm).

Use with care: S07 wind after 26 March. The new anemometer reads about a third of the true speed. We scaled it up (roughly ×3.5–4), but that is an estimate: fine for "windy or calm", not for wind records.

S01 Saddle Ridge Hut sits 1,000 m above the rest. It is colder, windier and lower in pressure for real, so report it separately from the network average. Most of its "rain" was snow.

EVERY STATION, EVERY SENSOR

STATION	TEMPERATURE	HUMIDITY	PRESSURE	WIND	RAIN	SUNLIGHT
S01 Saddle Ridge Hut · 2,913 m	●	●	●	●	○ mostly snow; totals ok	●
S02 North Mesa School	●	●	○ −6.5 hPa from 21 Mar (new board)	●	●	●
S03 Valley Library	○ 18–21 Mar removed (froze), 4 bad readings	●	●	●	●	●
S04 Foothills Comm. Garden	◆ highs too warm from late Mar; lows ok	●	●	●	◆ 23 Mar–2 Apr days wrong; total ok	●
S05 Eastside Rec Centre	●	○ 7–11 Mar removed (stuck at 100%)	●	●	●	●
S06 South Hills Fire Station	○ all sensors good; after the 24 Mar battery swap the clock ran on local time (6 h behind). Times corrected using the GPS.					
S07 Riverbend Farm	●	●	●	○ from 26 Mar ×3.5–4, estimate	●	●
S08 Plains Grange Hall	○ all sensors good; 36 hours missing 16–18 Mar (SIM ran out); 5 April arrived twice, copy removed so rain isn't counted double					

● good as is (dot only) ○ usable, already repaired or with a short gap ◆ do not use for the summary

SMALL JOBS THAT DON'T NEED THE TECHNICIAN

- S06 Ask the fire station to set the logger clock to **UTC**, not local time. Every new reading is still 6 hours off.
- S07 Ask the farm to photograph the new anemometer's label. The model number gives the exact factor to replace our estimate.
- S03 Order and post a spare temperature probe to the library.
- S08 Put the SIM on automatic top-up. S02 Calibrate the new pressure board when the kit is free.

PAPERWORK, AND THINGS THAT ARE FINE

- S05 The install sheet has the wrong latitude (39.570). The station's own GPS says **39.754**, about 20 km further north. Please correct the sheet.
- S02 About 3% of GPS messages arrive garbled (under 1% elsewhere). Weather data unaffected; worth a look at the GPS cable some day.
- **Looked odd but is real weather:** the 29 March storm (10–20 mm at 6 of 8 stations) and the fog over the whole network early on 30 March. Keep both.

Method: each station checked against its neighbours, its own GPS and the position of the sun. All 19 findings, with dates and evidence: **flags.csv**

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